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IPC-9712

Requirements for Automated Inspection Process Control for IC Substrates

If a conflict occurs between the English language and translated versions of this document, the English version will take precedence.

Developed by the Automated Inspection for IC Substrates Process Control Standard Task Group (7-25b) of the Process Control Management Committee (7-20) of IPC

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IPC-9712

Requirements for Automated Inspection Process Control for IC Substrates

1 SCOPE

This standard provides requirements for automated inspection systems — automated optical inspection (AOI), automated visual inspection (AVI) and metrology — to define, set-up, establish, and apply process control for manufacturing integrated chip substrates, including general and specific process and equipment conditions. Requirements include those for operating and inspection parameters, vision systems, lighting conditions, calibration, detectability, resolution, threshold limits and process windows, program setups, measurement system analysis (MSA), maintenance and verification protocols in addition to the generic requirements of IPC-9711.

1.1 Purpose The purpose of this standard is to set industry-defined requirements for AOI systems to reduce false calls, improve throughput and shorten cycle times to ensure quality and reliability of integrated chip substrates, including both in-line and outgoing inspection. This standard will also support electronics manufacturers to enable advanced manufacturing real-time data analytics and control capabilities.

1.2 Classification Classification **shall** be in accordance with IPC-9711.

1.3 Definition of Requirements Definition of requirements **shall** be in accordance with IPC-9711.

1.4 Process Control Requirements Process control **shall** be in accordance with IPC-9711.

1.5 Order of Precedence Order of precedence **shall** be in accordance with IPC-9711.

1.6 Terms and Definitions Other than those terms listed below, the definitions of terms used in this standard are in accordance with IPC-T-50.

1.6.1 Call System-generated identification of a potential defect at an inspection location or parameter.

1.6.2 False Call A call that has been dispositioned to be an acceptable condition.

1.6.3 Manufacturer The individual, organization or company responsible for the assembly process and verification operations necessary to ensure full compliance of assemblies to this standard.

1.6.4 Process Control A system or method to continually guide an operation in reducing variation in the processes or products to meet or exceed the goal in quality and performance.

1.6.5 Repair The act of reprocessing noncomplying product in a manner that assures compliance of the product with applicable drawings or specifications.

1.6.6 Supplier The individual, organization or company responsible for the manufacturing and delivery of the AOI system.

1.6.7 User The individual, organization, company, contractually designated authority or agency responsible for the procurement or design of electrical/electronic hardware and having the authority to define the Product Class and any variation or restrictions to the requirements of this standard (i.e., the originator/custodian of the contract detailing these requirements).

1.7 Facilities Cleanliness Facilities cleanliness **shall** be in accordance with IPC-9711.

1.8 Acceptance Requirements Acceptance requirements **shall** be in accordance with IPC-9711.

1.9 Abbreviations and Acronyms See Appendix D for full spellings of abbreviations (including elements) and acronyms used in this standard.

1.10 Inspection System Personnel Proficiency Inspection system evaluation review should be carried out by certified personnel.